Hardware Status and Build Guide

Disaster Communication and Resource Management System — what you own, what is missing, and how to wire it

Updated 12 Oct 2026 from your kit list. Scope: **one sensor node and one gateway**. Prices are rough Indian hobby-store ranges, not quotes. The wiring comes from `firmware/include/config.h` and has **not** been tested on real hardware yet.

1. Your kit against the build

Qty	Part you have	Role	Status
2	ESP32 DevKit (WROOM-32, 30/38-pin)	Node + gateway	OK
2	**Ra-02 LoRa module (SX1278, 433 MHz)** on blue breakout, **header strips loose in the bag**	Node + gateway radio	Works at 433 MHz. **Headers still need soldering** (16 joints each). Not the 868 MHz RFM95W originally planned
2	SMA whip antenna + u.FL pigtail	Radio antenna	OK. Check it is a 433 MHz antenna. Fit before power
2	USB data cable	Power + programming	OK
1	HC-SR04 ultrasonic sensor	Water level	Works, but **not waterproof**: indoor demo only
1	FC-37 / YL-83 rain plate + board	Rain	OK (wet/dry level only)
1	KY-026 IR flame sensor	Flame	OK (about 1 m range)
1	DS18B20 waterproof probe	Temperature	OK
2	830-point breadboard	Node + gateway	OK
1 pack each	Male-female and male-male jumper wires (40)	Wiring	OK
2 each	1 kΩ, 2 kΩ, 4.7 kΩ resistors	Dividers, DS18B20 pull-up	OK. Exactly enough
2	100–470 µF electrolytic capacitor	3.3 V rail for the radios	OK

2. What is still missing

Item	Needed?	Approx. price	Note
MQ-2 smoke sensor module	**Yes, for smoke detection**	₹150–250	Without it, run with GPIO 34 tied to GND (see section 10)
0.1 µF ceramic capacitors (5+)	Optional	₹10–20	Decoupling near the radio
Digital multimeter	Optional, strongly helpful	₹400–800	You can build without it by following the safety steps below
Soldering of the Ra-02 headers	**Yes**	free–₹50 at a shop	2 modules × 16 pins. Ask a shop or borrow an iron
Waterproof ultrasonic (JSN-SR04T)	Only for outdoor use	₹350–600	HC-SR04 is fine on a desk
5 V USB wall adapter	Optional	₹200–300	To run the node without the laptop

Not needed: extra resistors (1k + 2k pairs replace 10k + 20k), female-female jumpers, pin headers.

3. Cost to complete

Scenario	Cost
Minimum to get every sensor working (MQ-2 only)	₹150–250
Plus ceramic capacitors and a multimeter	₹560–1,070

4. Safe to build without a multimeter

- The one fatal mistake is 5 V on a 3.3 V GPIO. Only two signals are 5 V: the HC-SR04 ECHO and, later, the MQ-2 analog output. Both go through a 1 kΩ + 2 kΩ divider.
- Wire everything with USB **unplugged**. Plug in last. Recheck each wire against the table in section 10.
- Power the rain plate, flame sensor, DS18B20 and radio from **3V3**. Only the HC-SR04 (and the MQ-2) go on **VIN**.
- Tell resistors apart by colour bands: 1 kΩ is brown-black-red, 2 kΩ is red-black-red, 4.7 kΩ is yellow-violet-red.
- Fit the antenna before the radio is powered.
- After powering, open the Serial Monitor at 115200 baud. A JSON line prints every 10 s. A plausible `level_cm` shows the ECHO path works, and `rain` rises when the plate is wet.

Sensor choices and alternatives

Sections 5–9 cover each sensor: the part selected, then the alternatives with the trade-off.

5. Water level (flood)

You have: HC-SR04 (about 2 cm to 400 cm). It uses the same code and pins as the JSN-SR04T and suits a desk demo. It is not waterproof, so keep it above the water and dry. **Field upgrade: JSN-SR04T**, a sealed probe with about 25 cm to 450 cm range.

Alternative	Advantage	Drawback
JSN-SR04T	Waterproof probe, same code and pins	₹350–600; dead zone below about 25 cm
A02YYUW (UART waterproof ultrasonic)	Waterproof, 3.3 V-friendly, short dead zone (about 3 cm)	UART instead of pulse; slightly different code
Float switch	Very cheap, no calibration, works as a hard "level reached" trigger	One threshold per switch; no continuous reading
Submersible pressure transducer (4–20 mA or 0–5 V)	Accurate depth in the water, good for real deployments	₹2,000+, needs an ADC/amplifier stage and calibration
Exposed-trace "water level strip"	Almost free	Corrodes in days. **Not recommended.**

Notes: HC-SR04 echo is 5 V, so use the 1 kΩ + 2 kΩ divider. FLOOD triggers at `level_cm ⩾ 50`, and level is `MOUNT_CM - distance`. For a desk demo set `MOUNT_CM` to about 60 and fix the sensor 60 cm above the base, or lower the threshold in the server.

6. Rainfall

Selected: FC-37 / YL-83 rain plate. Reads wetness only (how wet the plate is), which is enough to show rain on the prototype.

Alternative	Advantage	Drawback
Tipping-bucket rain gauge (reed switch)	Real rainfall in mm per tip, standard in weather stations	₹1,200–2,500; needs an interrupt/pulse counter and mounting
Capacitive rain sensor	No exposed traces, so no corrosion	Less common, usually a bit pricier
Optical (infrared) rain sensor	No moving parts, detects drops	Costly; mostly used on vehicles and industrial setups

Notes: the rain plate corrodes and is wet/dry only, so it cannot give rainfall rate. For a credible "heavy rain" alert, upgrade to the tipping bucket and send mm per interval.

7. Flame (fire)

Selected: KY-026 IR flame sensor. A phototransistor tuned to the infrared band (about 760–1100 nm) emitted by flames. It has a digital alarm output with an adjustable threshold, plus an analog output.

Alternative	Advantage	Drawback
Multi-channel IR flame sensor (5-way)	Wider field of view, can estimate direction	Slightly more cost and wiring
UV flame sensor	Ignores sunlight, fewer false positives outdoors	Expensive; narrow supplier choices
MLX90640 thermal camera	Sees heat patterns over a whole area	₹4,000+; heavy processing for an ESP32
Temperature + smoke fusion (no flame sensor)	No extra part; confirms fire from two signals	Slower to react; no direct flame detection

Notes: range is only about 1 m, and direct sunlight causes false positives. Treat it as a demo-grade sensor; for real use, combine it with smoke and temperature before raising a FIRE alert.

8. Smoke and gas

To buy: MQ-2. A heated metal-oxide sensor that responds to smoke, LPG, methane, hydrogen and alcohol. Gives an analog level you threshold in firmware.

Alternative	Advantage	Drawback
PMS5003 / PMSA003 (laser particle sensor)	Measures real smoke particles (PM2.5/PM10)	₹1,500–2,500; UART, 5 V supply
MQ-7 / MQ-9	Detects carbon monoxide, which matters in fires	Same heater, drift and burn-in issues as MQ-2
MQ-135	Air-quality trend	Not a smoke detector; poor for fire
SGP30 / CCS811 / ENS160	Digital I2C, low power, TVOC/air-quality	Not designed as smoke detectors
Commercial photoelectric smoke detector with relay output	Certified detection	Extra wiring, bulky, needs its own 9 V supply

Notes: you do not own one yet; until it arrives, tie GPIO 34 to GND. MQ-2 needs a 24–48 hour burn-in, draws about 150 mA for its heater, and drifts with humidity. Its output can reach 5 V, so divide it down for the ESP32. It is not a certified smoke detector.

9. Temperature

Selected: DS18B20 waterproof probe. Digital 1-Wire sensor, −55 to 125 °C, ±0.5 °C, sealed in a steel probe. Needs a 4.7 kΩ pull-up.

Alternative	Advantage	Drawback
DHT22 (AM2302)	Temperature and humidity in one part	Slow (about 0.5 Hz), not waterproof
BME280 (I2C)	Temperature, humidity and pressure; falling pressure helps storm warning	Not waterproof; about ±1 °C
SHT31 / SHT40	More accurate humidity	Pricier; not waterproof
K-type thermocouple + MAX31855	Measures fire-level heat (hundreds of °C)	Extra amplifier board; about ₹500–800
DHT11	Cheapest	Poor accuracy and range. **Not recommended.**

10. Node wiring (from config.h, untested on hardware)

Device pin	ESP32 pin	Note
RFM95W 3.3V / GND	3V3 / GND	3.3 V only. **Never connect to 5 V.** Fit the antenna first
RFM95W NSS (CS)	GPIO 5	
RFM95W RST	GPIO 14	
RFM95W DIO0	GPIO 26	
RFM95W SCK / MISO / MOSI	GPIO 18 / 19 / 23	Default SPI pins
HC-SR04 VCC / GND	VIN (5 V) / GND	
HC-SR04 TRIG	GPIO 25	
HC-SR04 ECHO	GPIO 33	**Through a divider:** ECHO → 1 kΩ → GPIO 33 → 2 kΩ → GND
KY-026 VCC / GND / DO	3V3 / GND / GPIO 27	Powered at 3.3 V so its output is safe
No MQ-2 yet: GPIO 34	GND	Jumper GPIO 34 to GND so it reads 0 and raises no false SMOKE alerts. Remove when the MQ-2 is fitted
MQ-2 VCC / GND (when bought)	VIN (5 V) / GND	Heater needs 5 V
MQ-2 AO (when bought)	GPIO 34	**Through a divider:** AO → 1 kΩ → GPIO 34 → 2 kΩ → GND (second pair)
Rain plate board VCC / GND / AO	3V3 / GND / GPIO 35	
DS18B20 red / black	3V3 / GND	
DS18B20 yellow (data)	GPIO 4	4.7 kΩ resistor between data and 3V3

Notes:

- The gateway only needs the RFM95W wired as in the first five rows, plus the ESP32 on USB.
- Total draw at peak is roughly 500 mA (MQ-2 heater, LoRa transmit, ESP32 Wi-Fi). Use a good USB cable; weak ones cause random resets.
- GPIO 34 and 35 are input-only, which is fine for these analog reads.
- Drivers: Windows needs the CP210x or CH340 USB driver. On Linux, add your user to the `uucp` group.

11. Next steps and what to avoid

- **Demo build:** your current kit plus an MQ-2, one node and one gateway.
- **More credible field node:** swap the rain plate for a tipping bucket, add a BME280 for pressure, and require flame + smoke + temperature to agree before a FIRE alert.
- **Not recommended:** the exposed water-level strip, the HC-SR04 outdoors, DHT11, the 433 MHz Ra-02 for India, and treating MQ-2 as a certified smoke detector.